RtHDDump Content Difference Report

Executive summary (direct answers)

- WIDs do not change across Windows profiles. They identify hardware pins (speaker, headphone, mic, SPDIF) and stay constant across preferred device, Dolby, and enhancement states.
- Only one WID is overridden by the Windows driver: WID `0x1D` (`Drv=411111F0` vs `Codec=40471A6D`). Linux keeps the codec default (`Pin Default=40471A6D`).
- Windows coefficient changes are limited to headset plug: Wid `0x20` indices `0x10`, `0x46`, `0x67` toggle when the headset is plugged.
- Linux coefficient changes are limited to headset plug + automute/dualstream: Node `0x20` coefficient `0x46` toggles with headset plug; `0x77/0x78` move with automute/dualstream.
- Dolby/enhancement do not appear in WIDs or coefficients; they are encoded in the `REG_*` registry keys in Windows dumps.

The remainder of this report reconstructs the investigation into Windows internals, Linux internals, and a cross-platform investigation that maps each WID and coefficient to its inferred control meaning.

Windows internal investigation

WID meaning list (Windows WID → Linux Node mapping)

Windows `Wid=0x??` entries correspond directly to Linux `Node 0x??`. The table below lists each WID, its Linux node type, and the pin description (the practical meaning).

WID	Linux node type	Linux pin description	Win Codec	Win Drv	Drv!=Codec
0x12	Node 0x12 [Pin Complex] wcaps 0x40040b: Stereo Amp-In	[N/A] Line Out at Ext N/A	40000000	40000000	no
0x13	Node 0x13 [Pin Complex] wcaps 0x40040b: Stereo Amp-In	[N/A] Speaker at Ext Rear	411111F0	411111F0	no
0x14	Node 0x14 [Pin Complex] wcaps 0x40058d: Stereo Amp-Out	[Fixed] Speaker at Int N/A	90170120	90170120	no
0x17	Node 0x17 [Pin Complex] wcaps 0x40058d: Stereo Amp-Out	[Fixed] Speaker at Int N/A	90170120	90170120	no
0x19	Node 0x19 [Pin Complex] wcaps 0x40048b: Stereo Amp-In	[Jack] Mic at Ext Left	03A11030	03A11030	no
0x1A	Node 0x1a [Pin Complex] wcaps 0x40048b: Stereo Amp-In	[N/A] Speaker at Ext Rear	411111F0	411111F0	no
0x1B	Node 0x1b [Pin Complex] wcaps 0x40058f: Stereo Amp-In Amp-Out	[N/A] Speaker at Ext Rear	411111F0	411111F0	no
0x1D	Node 0x1d [Pin Complex] wcaps 0x400400: Mono	[N/A] SPDIF Out at Ext N/A	40471A6D	411111F0	yes
0x1E	Node 0x1e [Pin Complex] wcaps 0x400501: Stereo	[N/A] Speaker at Ext Rear	411111F0	411111F0	no
0x21	Node 0x21 [Pin Complex] wcaps 0x40058d: Stereo Amp-Out	[Jack] HP Out at Ext Left	03211010	03211010	no

Inference: WID `0x14/0x17` are internal speakers, WID `0x21` is the headphone jack, WID `0x19` is the external mic, and WID `0x1D` is SPDIF/aux. The only Windows driver override is WID `0x1D`.

Windows coefficient controls (Wid 0x20)

Windows stores vendor coefficients under Wid 0x20. The only coefficient changes tied to a Windows state are for headset plugged:

Index	Headset unplugged	Headset plugged	Inferred meaning
0x10	8A06	8B06	Headset plug toggle (Windows)
0x46	0004	0C34	Headset plug toggle (Windows)
0x67	1000	3000	Headset plug toggle (Windows)

Inference: headset plug is expressed via Wid 0x20 coefficient deltas, not via WID pin changes. Dolby/enhancement are not represented in Wid/coeff values and instead live in `REG_*` deltas.

Linux internal investigation

Linux node controls

Linux exposes the same pin controls as the Windows WID list above. The Linux pin defaults confirm the same roles: internal speakers on nodes `0x14/0x17`, headphone jack on `0x21`, mic on `0x19`, and SPDIF/aux on `0x1D`.

Linux coefficient controls (Node 0x20)

Linux exposes vendor coefficients under Node 0x20. Only three coefficients vary across Linux states:

State	Coeff 0x46	Coeff 0x77	Coeff 0x78	Inferred meaning
spk	0404	0050	00A6	Linux baseline
dual_h_spk	0C04	0063	0079	Headset plug toggles 0x46/0x77/0x78
dual_h_spk_no_automute	0C04	0023	0090	Automute toggle (0x77/0x78)
dual_h_spk_no_automute_dualstream	0C04	0046	005D	Dualstream toggle (0x77/0x78)

Inference:

- 0x46 is a headset-plug control in Linux.
- 0x77/0x78 encode automute/dualstream routing choices (Linux-specific behavior).

Cross investigation (Windows ↔ Linux)

Coefficient meaning list (merged view)

The table below lists every coefficient index that differs across Windows/Linux or changes with a state, and the inferred meaning based on observed deltas.

Index	Win base	Win headset	Linux base	Linux headset	Linux no_automute	Linux dualstream	Inferred meaning
0x03	F002	F002	0002	0002	0002	0002	Baseline mismatch (Win vs Linux)
0x04	AA09	AA09	AA89	AA89	AA89	AA89	Baseline mismatch (Win vs Linux)
0x08	4A37	4A37	4AB7	4AB7	4AB7	4AB7	Baseline mismatch

							(Win vs Linux)
0x10	8A06	8B06	8906	8906	8906	8906	Headset plug toggle (Windows)
0x1A	8C83	8C83	8003	8003	8003	8003	Baseline mismatch (Win vs Linux)
0x30	9007	9007	9004	9004	9004	9004	Baseline mismatch (Win vs Linux)
0x44	4900	4900	4500	4500	4500	4500	Baseline mismatch (Win vs Linux)
0x46	0004	0C34	0404	0C04	0C04	0C04	Headset plug toggle (Windows); headset plug toggle (Linux)
0x48	D049	D049	D011	D011	D011	D011	Baseline mismatch (Win vs Linux)
0x49	0049	0049	0045	0045	0045	0045	Baseline mismatch (Win vs Linux)
0x67	1000	3000	F000	F000	F000	F000	Headset plug toggle (Windows)
0x77	0000	0000	0050	0063	0023	0046	Headset plug + automute/ dualstream (Linux)
0x78	0000	0000	00A6	0079	0090	005D	Headset plug + automute/ dualstream (Linux)

Inconsistency list (what blocks matching Windows on Linux)

Inconsistency	Windows behavior	Linux behavior	Reproduction knob (Linux)
Pin config (WID 0x1D)	Driver overrides to `Drv=411111F0`	Linux keeps codec default `Pin Default=40471A6D`	Retask pin 0x1D to `0x411111F0` (firmware patch / hda-verb)

Vendor coeff baseline	Wid 0x20 values differ at indices `0x03/0x04/0x08/0x10/0x1A/0x30/0x44/0x46/0x48/0x49/0x67/0x77/0x78`	Node 0x20 coefficients keep Linux defaults	Write Node 0x20 coefficients to the Windows values
Headset plug handling	Wid 0x20 indices `0x10/0x46/0x67` toggle on plug	Linux uses pin mutes + coeff `0x46`	Apply Windows index values on plug/unplug and avoid automute pin mutes
Automute/dualstream	Not represented in Wid 0x20	Linux uses coeff `0x77/0x78` + pin amp-out	Set coeffs `0x77/0x78` to Windows baseline (0000) and manage routing explicitly

Direct reproduction path (Linux → Windows behavior)

1. Retask pin 0x1D to the Windows driver value 0x411111F0.
2. Apply the Windows Wid 0x20 coefficient baseline to Linux Node 0x20 (indices listed above).
3. On headset plug/unplug, mirror the Windows headset-plug deltas (0x10/0x46/0x67).
4. Neutralize Linux-specific automute behavior by setting 0x77/0x78 to Windows baseline (0000) and controlling routing explicitly.

Appendix: Regeneration

Generate the CSV summary and the PDF report from the current dumps:

```
python rthddump_report.py --output rthddump_summary.csv
typst compile RtHDDump_Report.typ RtHDDump_Report.pdf
```